PANDAS in 5 minutes

Reading files read file country.csv encoding='ISO-8859-1'

```
tent/sample_data/country.csv', encoding='ISO-8859-1')
ace un resumen cuantitativo
Con describe podremos devolver un resumen descriptivo con media, desviació
```

```
<bound method DataFrame.info of               Country      Capital Area(k
0           Afghanistan         Kabul    652 090            29.12
1               Algeria         Alger  2 381 741             34.3
2                Angola        Luanda  1 246 700            18.99
3   Antigua and Barbuda   Saint Johns        440             0.09
4             Argentina  Buenos Aires  2 766 890            40.09
..                  ...           ...        ...              ...
130      Western Sahara    No Capital    266 000             0.53
131             Vietnam         Hanoi    329 566            86.93
132               Yemen         San'a    531 869            22.49
133              Zambia        Lusaka    752 614            13.05
134            Zimbabwe        Harare    390 759            12.64

    Pop. Growth     Currency  Inflation  \
0         2.58%      Afghani  Not known
1         1.20%        Dinar  Not known
2         2.10%       Kwanza  Not known
3         1.30%  E.C. Dollar  Not known
4         1.05%         Peso  Not known
..          ...          ...        ...
130       2.83%    Not known  Not known
131       1.14%         Dong  Not known
132       2.79%         Rial  Not known
133       1.63%       Kwacha  Not known
134       1.53%       Dollar  Not known

                    Official name of Country
0               Islamic State of Afghanistan
1     People's Democratic Republic of Algeria
2                        Republic of Angola
3                                       NaN
4                        Argentine Republic
..                                      ...
130                                     NaN
131           Socialist Republic of Viet Nam
132                       Republic of Yemen
133                      Republic of Zambia
134                    Republic of Zimbabwe

[135 rows x 8 columns]>
            Country     Capital Area(km.sq) Population(mio) Pop. Growth  \
count           135         135         135             135         135
unique          135         133         133             124         107
```

```
top     Afghanistan  Basse Terre        340             0.1       2.75%
freq              1           2          2               4           3

         Currency  Inflation      Official name of Country
count         135        135                           112
unique         65          1                           112
top     CFA-Franc  Not known  Islamic State of Afghanistan
freq           13        135                             1
```

Accessing by row number (access to two fist rows)

```
df.iloc[0:2]
#Con iloc imprimo de la fila 0 hasta la 1 ya que la segunda no esta contenid
```

	Country	Capital	Area(km.sq)	Population(mio)	Pop. Growth	Currency	Inflat
0	Afghanistan	Kabul	652 090	29.12	2.58%	Afghani	Not kn
1	Algeria	Alger	2 381 741	34.3	1.20%	Dinar	Not kn

Selecting some columns (select only 3 columns, later only 3 columns of 2 first records)

```
df.iloc[0:2, 0:3]
#Con iloc imprimo de la fila 0 a la 1 y de la columna 0 hasta la 2
```

	Country	Capital	Area(km.sq)
0	Afghanistan	Kabul	652 090
1	Algeria	Alger	2 381 741

Filtering by condition. Apply a filter to a column

```
df['Capital'] == 'Alger'
#Para filtrar elegimos una columna del df y ponemos un comparador == para co
#En este caso la Capital Alger
```

	Capital
0	False
1	True
2	False
3	False
4	False
...	...
130	False
131	False
132	False
133	False
134	False

135 rows × 1 columns

dtype: bool

Add new column with default value

```
df["NuevaColumna"] = "Hola mundo"
#Para crearnos una nueva columna es muy parecido al ej anterior solo que 
#Asignamos el nombre que queremos para nuestra columna y le asignamos val
df.head() #Mostramos las primeras filas
#print(df)
```

Country Capital Area(km.sq) Population(mio) Pop. Growth Currency Inflat

Add new column with calculated value

```
#El valor calculado será la concatenacion de pais y capital
df['Resumen'] = df['Country'] + '-' + df['Capital']
df.head() #Mostramos las primeras filas
```

1 Algeria Alger 2 381 741 34.3 1.20% Dinar Not kn

	Country	Capital	Area(km.sq)	Population(mio)	Pop. Growth	Currency	Inflat
0	Afghanistan	Kabul	652 090	29.12	2.58%	Afghani	Not kn
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3	Antigua and Barbuda	Saint Johns	440	0.09	1.30%	E.C. Dollar	Not kn
4	Argentina	Buenos Aires	2 766 890	40.09	1.05%	Peso	Not kn

3 and Barbuda Johns 440 0.09 1.30% Dollar Not kn
1 Algeria Alger 2 381 741 34.3 1.20% Dinar Not kn
4 Argentina Buenos Aires 2 766 890 40.09 1.05% Peso Not kn

Remove column

```
df.drop(columns=["NuevaColumna"], inplace=True)
#Con inplace true nos aseguramos de eliminar la columnal df original
df.head()#Mostramos las primeras filas
```

	Country	Capital	Area(km.sq)	Population(mio)	Pop. Growth	Currency	Inflat
0	Afghanistan	Kabul	652 090	29.12	2.58%	Afghani	Not kn
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2	Angola	Luanda	1 246 700	18.99	2.10%	Kwanza	Not kn
3	Antigua and Barbuda	Saint Johns	440	0.09	1.30%	E.C. Dollar	Not kn
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Writing files with PANDAS. save dataframe in a csv file

```
#A continuación con el siguiente comando me guardaré el csv
df.to_csv("country_edit.csv", sep= "|" ,index=False)
#Por defecto se guardará en la carpeta archivos
#sep será el separador que será mostrado entre las columnas
#Index=False hará que no se guarde la columna de indices
```